

TRAINING MANUAL

Cross-Asset Ratio Framework

How to gauge economic performance, mark to market multi-asset exposures, and read relative value before and after macro shocks

A practical guide for beginner, intermediate, and expert traders

Ratio map: what each pair tends to diagnose

Use this as a first-pass map, then confirm with event context and related ratios.



Figure 1. A ratio map: each pair is most useful when it is tied to a clear economic question.

Core idea: a ratio does not tell you what is expensive in absolute terms. It tells you which narrative is winning right now.

Executive TL;DR

What this guide trains you to do

- Use ratios to compare two competing economic stories rather than staring at one outright chart.
- Separate clean beta from business-model leverage, operating leverage, and crisis hedges.
- Mark to market not only price, but the message of price: growth, fear, inflation shock, or normalization.
- Quantify moves with raw ratios, log spreads, event-window returns, and rolling z-scores instead of eyeballing alone.
- Read geopolitical events in sequence: pre-event positioning, first reaction, first weekly close, and 2-6 week follow-through.
- Build a small ratio dashboard that can guide allocations across growth assets, defensives, commodities, and crypto proxies.

How to use this manual

Read the guide in layers. The beginner layer teaches what a ratio is and how to interpret a rising or falling line. The intermediate layer adds normalization, z-scores, event studies, and confirmation logic. The expert layer adds residual analysis, hedge ratios, volatility scaling, and anomaly detection.

Throughout the guide, the goal is not to predict the economy with one magical indicator. The goal is to build a repeatable decision process: identify the question, choose the right pair, quantify the move, confirm the regime, and then decide whether your allocation should lean toward cleaner beta, defensives, or leveraged proxies.

Five-step mark-to-market workflow

Use the same sequence every day to keep interpretation consistent.

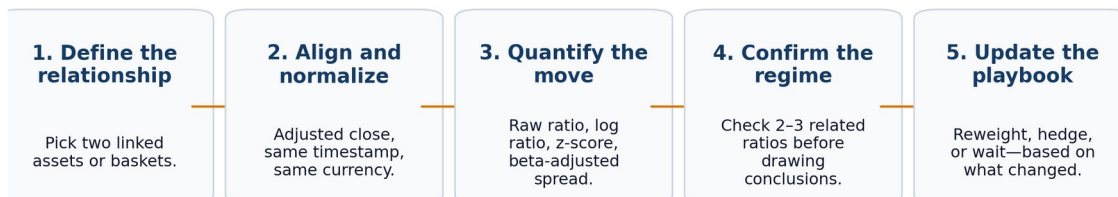


Figure 2. The same five-step process should be applied to every ratio so interpretation stays consistent.

Quick starter watchlist

- Copper/Gold or SMH/XLU for growth versus safety.
- HYG/IEF or IWM/XLP for risk appetite and credit transmission.
- USO/JETS or XLE/JETS for inflation shock and demand destruction.
- SLV/SMH when you want to compare a dual-use hard asset with pure tech leadership.
- COIN/IBIT, MSTR/IBIT, and MARA/IBIT when you want to know whether the market prefers clean Bitcoin beta or layered crypto leverage.

1. Why ratios work

Every asset price mixes several forces at once: macro growth, liquidity, policy expectations, positioning, volatility, and idiosyncratic noise. A ratio strips part of that noise away by forcing capital to choose between two narratives. If Asset A divided by Asset B is rising, A is outperforming B. That does not automatically mean A is healthy in absolute terms - both legs can fall while the ratio still rises - so good traders read the ratio and both legs together.

Ratios are most powerful when both legs are liquid, widely followed, and economically related. For example, oil versus airlines has a direct cost link; semiconductors versus utilities is a clean growth versus defense expression; Bitcoin spot ETF versus exchange or miner equities separates clean underlying beta from monetization and leverage.

How to read a market ratio

A ratio converts two noisy prices into one relative signal.

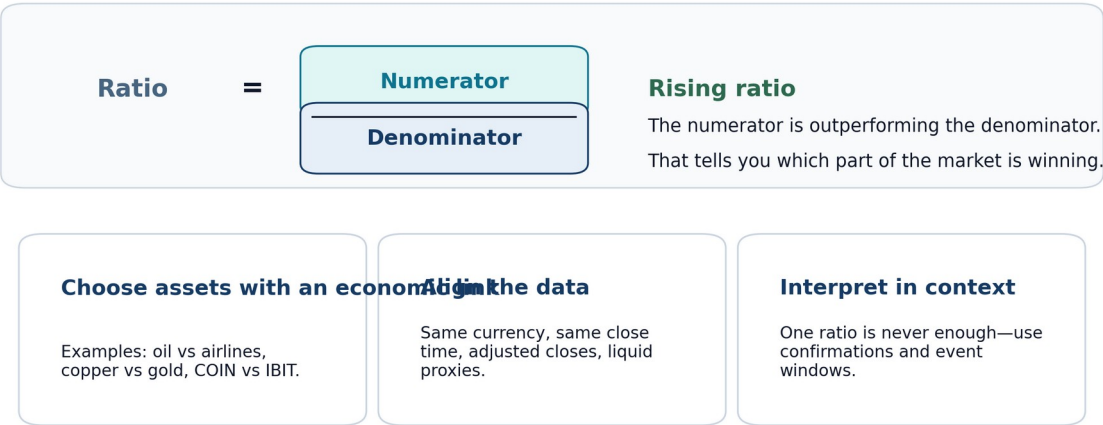


Figure 3. A market ratio is simply two prices converted into one relative signal.

What makes a good ratio

Design test	What to ask
Clear question	The pair must answer one question at a time: growth vs defense, inflation vs demand, or clean beta vs leverage.
Economic linkage	There should be a believable causal story connecting the two legs, not just a chart that happened to look good in the past.
Comparable structure	Use assets in the same currency, with similar trading hours when possible, and with transparent price formation.
Liquidity	Thin products generate false signals. Use deep futures, large ETFs, or liquid large-cap equities when possible.
Confirmability	A good ratio can be cross-checked with at least one or two independent confirmers. If it cannot be verified, it should not

Design test	What to ask
	drive size.

Beginner: The simplest framing

- Ask: what should win if my macro view is right?
- Put that asset in the numerator; put the opposing expression in the denominator.
- Do not use a pair unless you can explain it in one sentence before you look at the chart.

Intermediate: Improve signal quality

- Use adjusted closes and a shared closing time when possible.
- Track the ratio on daily and weekly closes so you can separate noise from follow-through.
- Always compare the ratio move with the move in each leg.

Expert: Reduce structural noise

- Use log spreads, rolling betas, and residuals when the simple ratio is dominated by one broad factor.
- Rank the same signal across multiple comparable pairs rather than betting on one chart in isolation.
- Monitor ETF structure, futures roll yield, funding conditions, and issuance/dilution risk when proxies are used.

2. Quantify the ratio, not just the story

A compelling macro story is not enough. The ratio has to be measured relative to its own history, volatility, and event context. The same 5 percent move can be meaningless in a highly volatile crypto proxy and highly meaningful in a slow-moving credit ratio.

Quantify the ratio, not just the story

The same move can mean very different things depending on volatility and history.

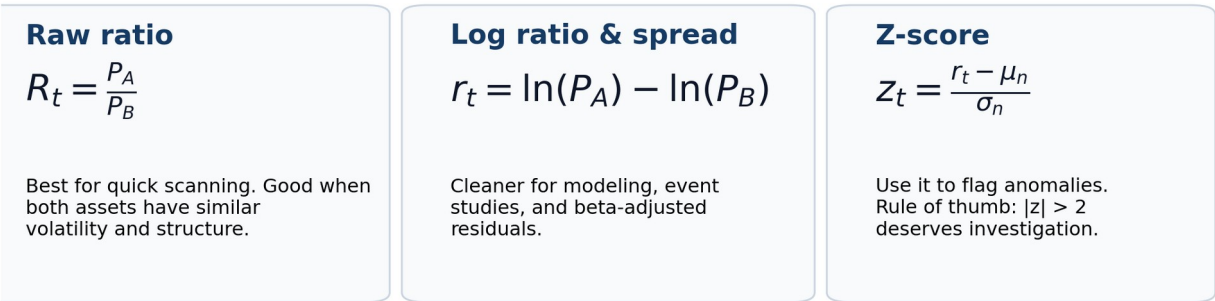


Figure 4. The same move means different things depending on the normalization you use.

Core calculations

Metric	Formula	Why it is useful
Raw ratio	$R_t = \text{Price}_A / \text{Price}_B$	Fast, intuitive, useful on charts and for event-window returns.
Log spread	$s_t = \ln(\text{Price}_A) - \ln(\text{Price}_B)$	Cleaner for statistics, residuals, and mean-reversion work.
Z-score	$z_t = (s_t - \text{mean}_n) / \text{stdev}_n$	Shows how unusual the current spread is relative to its own recent history.
Event return	$ER = R_t / R_{\text{event}} - 1$	Measures how the ratio behaved from the event date to now.
Beta-adjusted residual	$\text{eps}_t = r_{A,t} - \text{beta} * r_{B,t}$	Useful when you want to isolate asset-specific outperformance from broad market beta.

Practical thresholds

Approximate move	Working interpretation
0.5 sigma or less	Often normal noise unless multiple confirmers say otherwise.
1.0 to 1.5 sigma	Start investigating. This is often where regimes begin to separate.
2.0 sigma	Meaningful. The move deserves explanation, confirmation, and often a position review.
3.0 sigma or more	Treat as a dislocation or regime event until proven

Approximate move	Working interpretation
	otherwise.

Data hygiene checklist

- Use the same currency. Do not compare a USD asset with a local-currency asset unless FX is part of the thesis.
- Align trading windows. Crypto trades 24/7 while equities do not; event readings can differ depending on the cut-off.
- Use adjusted data for equities and ETFs when splits or distributions matter.
- Know the proxy. USO tracks futures, not barrels in a tank; miner and exchange equities are not the same thing as spot BTC.
- Look at at least two windows: first reaction and first weekly close. One-day shocks often reverse.

A spreadsheet-ready formula stack

For most traders, a simple spreadsheet is enough. Store Date, Asset A close, Asset B close, Ratio, Log spread, 20-day average, 20-day standard deviation, and 20-day z-score. That gives you a baseline dashboard without overengineering.

```
=A_close / B_close
```

```
=LN(A_close) - LN(B_close)
```

```
=(current_spread - AVERAGE(last_20_spreads)) / STDEV.S(last_20_spreads)
```

```
=(current_ratio / event_ratio) - 1
```

3. A macro map for ratio traders

Before using any pair, place it on a macro map. Most ratios are trying to diagnose one of four broad conditions: growth/risk-on, fear/safety, inflation or stagflation shock, and relief/disinflation. The pair itself is less important than the question it answers.

Ratio map: what each pair tends to diagnose

Use this as a first-pass map, then confirm with event context and related ratios.



Figure 5. Ratios are easiest to read when each one is tied to a regime question rather than a vague view.

Examples: Copper/Gold and SMH/XLU usually improve when the market rewards growth and cyclical leadership. USO/JETS tends to rise when energy costs become a tax on growth. GLD/Copper tends to rise when the market prefers safety or real-rate hedges to industrial demand. SLV/SMH sits in a more ambiguous zone because silver carries both industrial utility and hard-asset/speculative demand.

One of the biggest mistakes in ratio trading

Treating a single ratio as a complete macro dashboard. The best traders use clusters: one primary ratio, one confirmer, and one risk-control ratio. If all three agree, conviction can rise. If they disagree, size should fall.

4. Case study: silver versus semiconductors

Silver is a special asset because it can act like an industrial input, a precious metal, and a speculative vehicle. The Silver Institute says industrial demand reached a record 680.5 million ounces in 2024, supported by record electronics and electrical demand tied to AI-related hardware. Semiconductors, by contrast, are a far cleaner growth and capex expression. The Semiconductor Industry Association says global semiconductor sales reached \$791.7 billion in 2025 and are projected to approach \$1 trillion in 2026. That makes SLV/SMH a useful question: is capital rewarding a dual-use hard asset, or is it rewarding pure tech leadership?

The geopolitical backdrop matters here. Reuters reported that the 2026 war shock began on February 28, when U.S. and Israeli strikes on Iran helped stall flows through the Strait of Hormuz and flipped the oil market outlook toward deficit. Earlier Iran-related episodes in April 2024 and June 2025 also produced the classic first-order reaction of higher oil, firmer gold, and lower risk appetite. So the silver/semiconductor ratio should never be read in isolation from oil, gold, and credit.

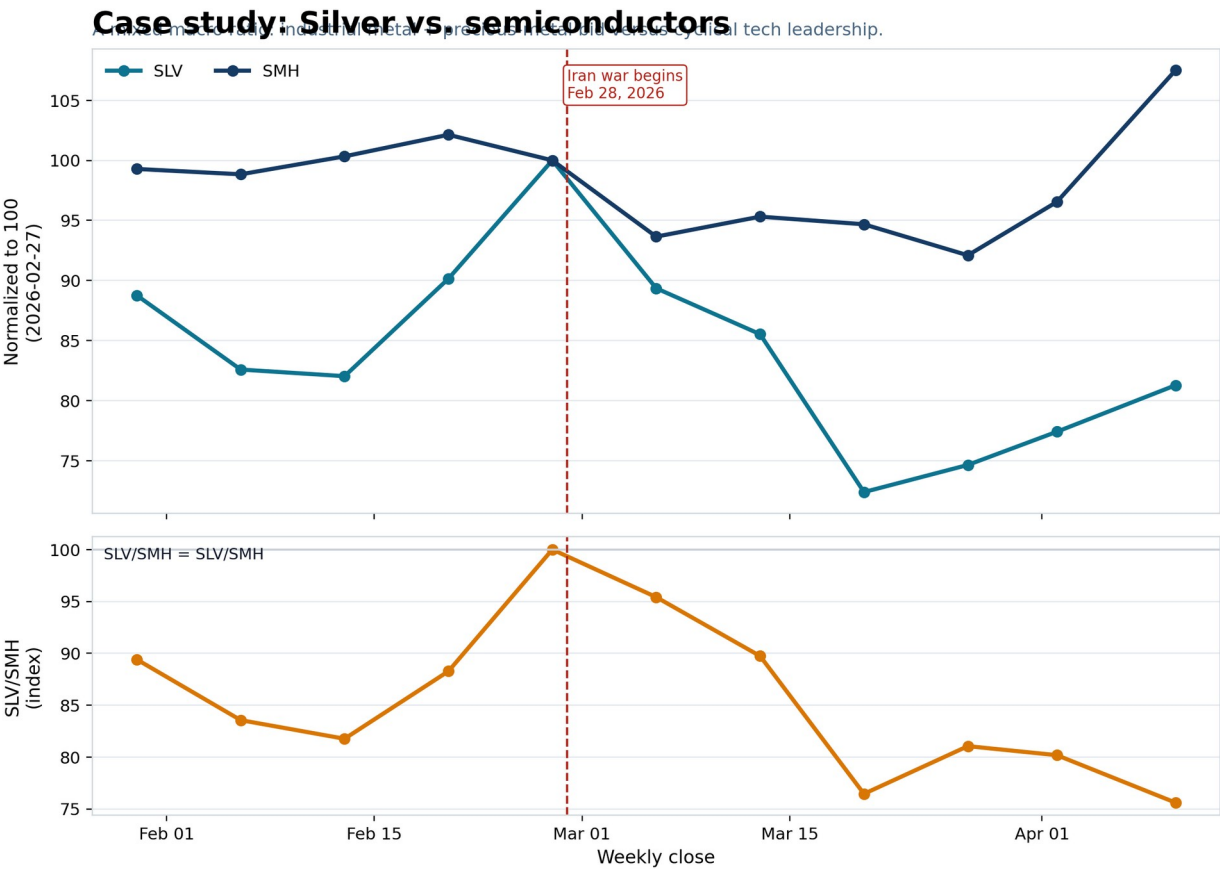


Figure 6. Weekly closes from Jan. 30 to Apr. 10, 2026. The vertical marker highlights the Feb. 28 war onset.

Window	SLV/SMH move	Working interpretation
Jan. 30 -> Feb. 27	+11.9%	Silver outperformed semiconductors into the geopolitical stress window.
Feb. 27 -> Mar. 2	-4.0%	Immediate post-event noise: the first session did not confirm a lasting silver breakout.
Feb. 27 -> Apr. 10	-24.4%	Semiconductors regained leadership over the following weeks; the silver bid

Window	SLV/SMH move	Working interpretation
		faded relative to growth tech.

How to read the signal

- If SLV/SMH rises together with Copper/Gold and HYG/IEF, the message is probably industrial growth or reflation rather than fear.
- If SLV/SMH rises while USO/JETS and GLD/Copper also rise, the message is more likely inflation shock, geopolitical stress, or a safety bid.
- If SLV/SMH rises only because SMH is collapsing while silver is flat, the ratio is warning about tech stress more than it is endorsing silver demand.

Beginner: A clean first-pass interpretation

- Rising SLV/SMH = silver is outperforming semiconductors.
- Ask whether that strength comes from silver rising, semiconductors falling, or both.
- Confirm with oil and gold before calling it an economic signal.

Intermediate: Separate growth from fear

- Compare SLV/SMH with Copper/Gold and HYG/IEF. Those pairs help tell you whether the market is rewarding industry or hiding in hard assets.
- Use both event-day and weekly-close windows. The first session after a geopolitical shock is often messy.
- Check silver-specific fundamentals such as solar and electronics demand before making the ratio purely macro.

Expert: Use a residual view

- Estimate silver alpha after controlling for gold and copper or broad commodity beta.
- If SLV/SMH rises and silver residuals are positive, the move is more likely silver-specific strength rather than a generic precious-metal bid.
- If the ratio breaks out but the residual stays flat, the signal may be denominator weakness masquerading as numerator strength.

5. Case study: oil versus airlines

USO/JETS is one of the most intuitive inflation-shock ratios in public markets. Oil is a direct source of fuel costs and inflation pressure. Airlines are cyclical businesses that benefit from travel demand but suffer when fuel costs spike and consumers face higher overall expenses. A rising oil-versus-airlines ratio often tells you the market is paying for cost pressure and demand destruction at the same time.

In the 2026 Iran-war window, this ratio sent a much cleaner message than SLV/SMH. From the last weekly close before the Feb. 28 war onset to Apr. 10, USO/JETS rose roughly 67.4 percent. That is exactly what you would expect when the market fears a sustained oil supply disruption.

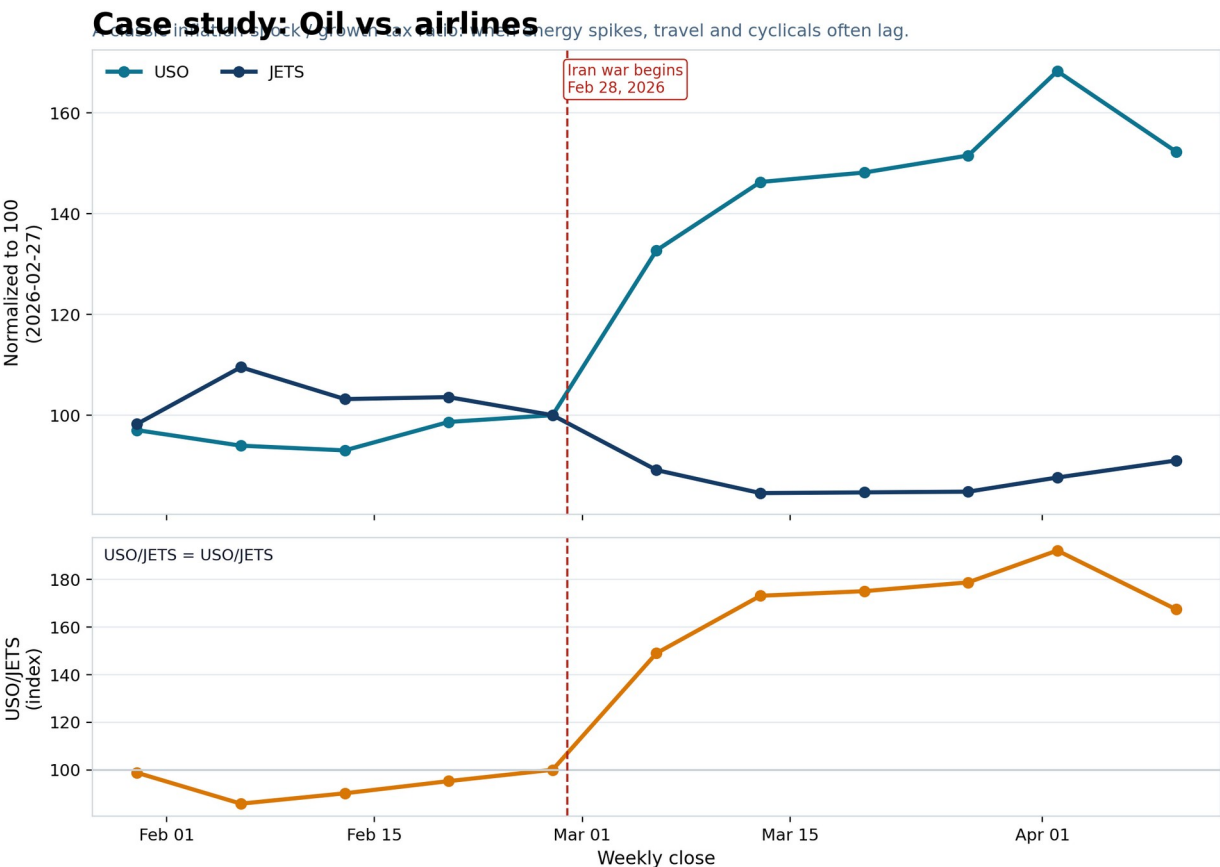


Figure 7. USO/JETS reacted as a classic shock ratio: oil up, airlines down, and the spread widened sharply.

Window	USO/JETS move	What it usually means
Feb. 27 -> Mar. 2	+9.3%	Immediate confirmation that the first-order response was an oil shock, not a benign growth scare.
Feb. 27 -> Apr. 10	+67.4%	Persistent input-cost pressure; a strong stagflation or shock signature.

What this ratio does well

- It captures both inflation pressure and the tax that energy imposes on discretionary demand.
- It often reacts faster than slow-moving macro data releases.
- It is a useful confirmer for other hard-asset ratios. If oil is surging and airlines are weakening, fear-based commodity strength deserves more respect.

Important caveat

USO is a futures-based ETF, so long-run readings can be distorted by roll yield and curve shape. For a secondary check, compare XLE/JETS or spot Brent versus a broad airline basket. The signal is similar, but the structure noise is different.

6. Case study: Bitcoin public-market proxies

A powerful way to think about ratios is the proxy stack. One underlying asset can have several listed expressions, and each one embeds a different combination of beta, leverage, financing, and business-model exposure. Bitcoin is an ideal example because it has a clean listed spot proxy, an exchange proxy, a leveraged balance-sheet proxy, and a mining proxy.

IBIT is close to clean BTC beta inside public markets. BlackRock lists one holding, uses the CME CF Bitcoin Reference Rate - New York Variant as benchmark, and reported \$57.67 billion of net assets with a 0.25 percent expense ratio as of Apr. 10, 2026. COIN is not spot BTC; it monetizes trading activity, custody, and platform usage. Coinbase said 2025 trading volume reached \$5.2 trillion, market share rose to 6.4 percent, and more than 12 percent of all crypto in the world resided on Coinbase in 2025. MSTR is a levered corporate treasury expression: Strategy disclosed holdings of roughly 762,099 BTC as of Mar. 29, 2026. MARA adds operating leverage through mining economics, energy costs, and network conditions.

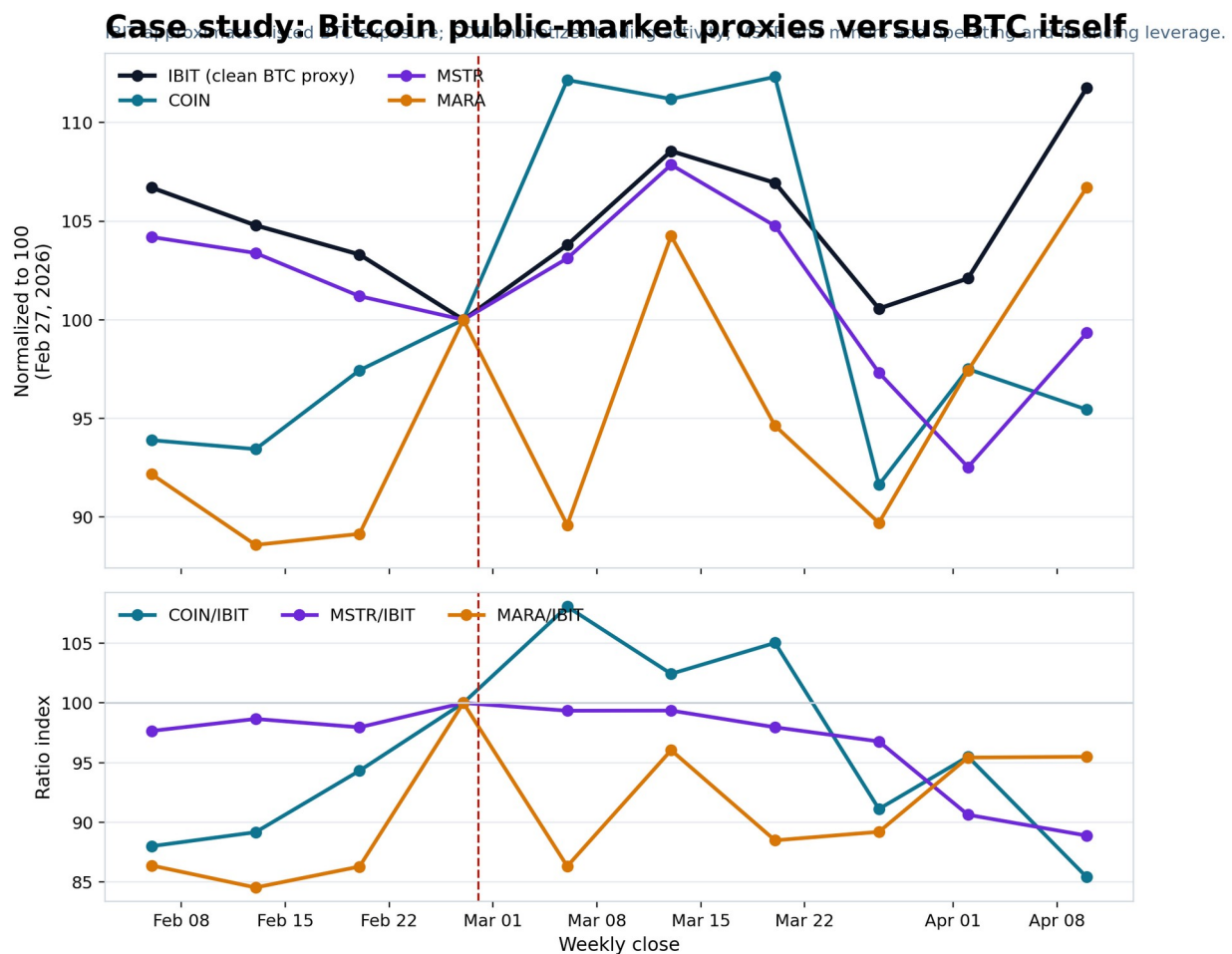


Figure 8. Public-market BTC proxies can diverge meaningfully from a listed spot ETF depending on leverage, monetization, and operating risk.

Pair	Feb. 27 -> Apr. 10	Primary read
COIN / IBIT	-14.6%	The market preferred cleaner BTC exposure to exchange monetization and activity risk.
MSTR / IBIT	-11.1%	Leverage premium compressed relative to spot-linked exposure.

Pair	Feb. 27 -> Apr. 10	Primary read
MARA / IBIT	-4.5%	Mining beta lagged clean spot exposure, though the move was less dramatic than COIN or MSTR.

How to use the proxy stack

- If IBIT is strong and COIN/IBIT is also rising, the market wants both the underlying and the monetization layer.
- If IBIT is strong but COIN/IBIT falls, the market wants clean BTC beta without taking exchange-specific or regulatory risk.
- If MSTR/IBIT rises too quickly, the leverage premium may be stretching beyond what spot BTC justifies.
- If MARA/IBIT collapses while BTC is stable, investigate hashprice, power costs, capital structure, or miner-specific issuance risk.

Beginner: Think in layers

- Spot ETF = cleaner Bitcoin beta.
- Exchange stock = volume, spreads, custody, and regulation.
- Treasury company = BTC plus financing and balance-sheet leverage.
- Miner = BTC plus operating leverage to hashprice and power costs.

Intermediate: Use relative ranking

- Rank COIN/IBIT, MSTR/IBIT, and MARA/IBIT together. When all three weaken against IBIT, the market is de-risking the layered expressions of BTC.
- When they strengthen together, speculative appetite and business-model confidence are broadening beyond spot.
- Compare immediate reaction versus 2-6 week follow-through; the first day can be noisy while the multi-week trend tells the real story.

Expert: Turn the stack into a diagnostic dashboard

- Measure each proxy's beta to IBIT or spot BTC and track the residual, not just the raw price.
- Add issuer-specific variables: exchange volume for COIN, financing or issuance for MSTR, and network difficulty or hashprice for miners.
- The cleanest signal often comes from the change in premium: is the market paying more or less for leverage and monetization relative to spot?

7. How to spot and quantify anomalous price action

An anomaly is not simply a move you did not expect. It is a move that is unusual relative to the ratio's own history, its confirming signals, or the economic logic linking the two legs. Your job is to decide whether the anomaly is a true regime change, an overshoot, or structure noise.

Anomaly type	What it looks like	What to do
Breakout without leg strength	The ratio makes a new high, but the numerator is barely rising and the denominator is just collapsing.	Treat as weaker quality. You may be seeing denominator stress, not clean numerator leadership.
Cross-confirmation failure	SLV/SMH rises, but oil, gold, and credit do not confirm the same narrative.	Reduce conviction until a second or third ratio agrees.
Correlation breakdown	A pair that usually moves together suddenly decouples.	Check for idiosyncratic news, ETF structure, regulation, or company-specific risk.
Time-window mismatch	Crypto and equities tell different stories around the same event.	Rebuild the study on a common cut-off and compare daily and weekly closes.
Premium compression	Leveraged or monetized proxies lag the clean underlying.	This often signals a preference for quality or a repricing of leverage, not weakness in the underlying itself.

A simple anomaly score

One practical scoring method

- Add 1 point if the absolute z-score is above 1.5.
- Add 1 point if the 5-day ratio move is more than 1.5 times its 20-day realized volatility.
- Add 1 point if the 60-day rolling correlation between the legs breaks sharply from its 6-month average.
- Add 1 point if at least two confirming ratios disagree with the signal.
- 0-1 points = routine noise, 2 points = watch list, 3 or more = investigate or reduce size.

Why anomalies matter for risk management

The point of anomaly detection is not to prove you are right. It is to protect you from overconfidence. If the ratio is screaming but the confirmation matrix is weak, size down. If the ratio is only unusual because the denominator had idiosyncratic bad news, do not mistake that for a broad economic shift.

8. Building a mark-to-market cross-asset dashboard

A good dashboard should tell you which sleeve of the market is winning capital, not just which ticker is green. The cleanest design is to split your book into a few economic sleeves and assign 2-3 diagnostic ratios to each. Then update the ratios every close and write one sentence on what changed.

Sleeve	Typical holdings	Diagnostic ratios	If the score rises
Growth / cyclical	SMH, IWM, HYG	SMH/XLU, Copper/Gold, HYG/IEF	Growth leadership and improving risk appetite.
Defensive / hard asset	GLD, Treasuries, silver	GLD/Copper, SLV/SMH	Safety demand or inflation hedging is winning.
Shock / input cost	Oil, energy, transports	USO/JETS, XLE/JETS	Input costs are becoming a tax on demand.
Crypto structure	IBIT, COIN, MSTR, MARA	COIN/IBIT, MSTR/IBIT, MARA/IBIT	The market is choosing between clean beta and layered leverage.

A simple daily routine

1. Update closes and the ratio table.
2. Mark each ratio as uptrend, downtrend, or range on daily and weekly time frames.
3. Check whether the move came from numerator strength, denominator weakness, or both.
4. Compare the move with one confirmer and one control ratio.
5. Write a one-line conclusion: what did the market pay for today?
6. Decide whether your allocation should prefer cleaner beta, defensive assets, or reduced size.

Position sizing reminder

Strong narratives can tempt traders to size too aggressively. A better default is volatility scaling. When several ratios agree, confidence can rise, but position size should still respect each pair's realized volatility and liquidity.

Volatility-scaled weight: $w_i = (1 / \sigma_i) / \text{SUM}(1 / \sigma)$

9. Playbooks by experience level

Beginner playbook

Start with five ratios only

- Copper/Gold for growth versus safety.
- HYG/IEF for credit risk appetite.
- USO/JETS for inflation shock and demand tax.
- SLV/SMH for dual-use hard asset versus pure tech leadership.
- COIN/IBIT for monetization risk versus clean Bitcoin beta.

- Use daily charts for awareness and weekly closes for conviction.
- Do not trade a ratio you cannot explain in one sentence.
- Always ask whether the numerator is actually strong or whether the denominator is simply weak.
- Wait for at least one confirming ratio before increasing size.
- Journal the signal, the confirmer, and what would invalidate the read.

Intermediate playbook

Add structure without overcomplicating it

- Track 20-day z-scores and event-window returns.
- Use a confirmation matrix: primary ratio, confirmer, and control ratio.
- Compare immediate event reaction with the first weekly close and the next 2-6 weeks.
- Keep a small set of proxy caveats next to each pair.

- Use a regime journal: growth, fear, inflation shock, or relief.
- Rank similar ratios together. For crypto, compare COIN/IBIT, MSTR/IBIT, and MARA/IBIT rather than staring at one chart.
- Review false signals. The best improvement often comes from studying where confirmation failed.

Expert playbook

Turn the dashboard into a research process

- Estimate rolling betas and residuals so you can separate true outperformance from market beta.
- Volatility-scale the book and use liquidity-adjusted limits for thin or event-driven pairs.
- Track distributional behavior after 2-3 sigma shocks: fade, follow, or wait for weekly confirmation.
- Map each proxy stack into clean beta, monetization layer, financing layer, and operating layer.

- When the ratio is strong but the residual is weak, you may be looking at denominator stress instead of numerator leadership.
- Use cross-sectional ranking to avoid falling in love with one narrative. Ask which pair is the cleanest expression of the same regime.
- Think like a risk manager: every ratio has a failure mode, and that failure mode should be written down before size goes up.

10. Tangible ratio library

The table below is designed as a practical watchlist. These are not rigid rules. They are recurring economic relationships that tend to carry information when the proxies are liquid and the broader context is understood.

Ratio	What it diagnoses	If the ratio rises	Best confirmer or caveat
Copper / Gold	Industrial growth versus hard-asset safety.	Growth, reflation, and cyclical are improving.	Confirm with SMH/XLU; China supply distortions can muddy the signal.
SMH / XLU	Tech and capex leadership versus bond-like defense.	Risk-on and lower fear.	Confirm with HYG/IEF; mega-cap concentration matters.
HYG / IEF	Credit spread appetite versus Treasury safety.	Risk appetite and easier financial conditions.	Rates can distort the read; use with credit spreads if available.
IWM / XLP	Domestic cyclical versus consumer staples.	Small-cap and household confidence are improving.	Balance-sheet sensitivity can dominate in rate shocks.
KRE / XLU	Banks versus defensives.	Better credit transmission and a friendlier growth backdrop.	Bank-specific stress can overpower macro.
XHB / TLT	Housing cyclical versus long-duration rate sensitivity.	Housing and rate-sensitive growth are improving.	Mortgage-rate moves dominate this pair.
USO / JETS	Oil shock versus travel demand.	Stagflation, input-cost pressure, or geopolitical stress.	USO structure matters; confirm with XLE/JETS.
SLV / SMH	Dual-use hard asset versus pure tech leadership.	Either industrial metal leadership or a hard-asset bid is gaining ground.	Confirm with Copper/Gold and GLD/Copper to separate growth from fear.
GLD / Copper	Safety and real-rate hedge versus industrial demand.	Fear, slowdown, or policy stress.	Gold is also sensitive to real yields and the dollar.

Ratio	What it diagnoses	If the ratio rises	Best confirmer or caveat
COIN / IBIT	Exchange monetization versus clean BTC beta.	The market is rewarding platform activity, fees, and product breadth more than spot exposure.	Check trading volume and options activity; regulation matters.
MSTR / IBIT	Leverage premium versus clean BTC beta.	The market is willing to pay up for balance-sheet leverage and optionality.	Monitor financing, issuance, and premium compression risk.
MARA / IBIT	Mining operating leverage versus clean BTC beta.	The market is rewarding high-beta miner exposure.	Check hashprice, power costs, and dilution risk.
BTC / Gold	Digital-risk hard-money beta versus classic hard-asset safety.	Liquidity and speculative appetite are broadening.	24/7 trading and macro policy shocks can create timing mismatches.

11. A daily and weekly worksheet

A framework only becomes useful when it is repeated. The worksheet below can be copied into a notebook or spreadsheet and used every day.

Field	What to write
Question	What am I trying to diagnose: growth, fear, inflation shock, or proxy quality?
Primary ratio	Which pair best answers that question?
Leg review	Did the move come from numerator strength, denominator weakness, or both?
Quant score	What are the event return, z-score, and weekly trend?
Confirmers	Which second and third signals agree or disagree?
Action	What changes in allocation, sizing, or conviction?

Five questions to ask before every trade

1. Why should this numerator beat this denominator if my thesis is right?
2. What would confirm that the move is macro rather than idiosyncratic?
3. What is the ratio's current z-score or event-window move relative to history?
4. What is the most likely false positive in this pair?
5. If I am wrong, which leg is most likely the source of the error?

TL;DR for the whole manual

- A ratio is a competition between two assets.
- The best ratios ask one question at a time and can be confirmed by other pairs.
- Do not trust a ratio until you understand both legs, the proxy structure, and the event window.
- In shocks, use sequence: pre-positioning, first reaction, first weekly close, and follow-through.
- Prefer cleaner beta when layered proxies start to underperform.

Appendix: source notes and data notes

The case studies use public-market proxies and are intended for education. Prices in the illustrations were based on weekly closes from Jan. 30 through Apr. 10, 2026 for SLV, SMH, USO, JETS, IBIT, COIN, MSTR, and MARA. The charts are meant to teach the framework, not to serve as trade recommendations.

Selected source notes

[A] Reuters, Apr. 10, 2026: analysts said the Iran war, which began on Feb. 28, had stalled flows through the Strait of Hormuz and flipped the 2026 oil market outlook toward deficit.

[B] Reuters, June 13, 2025: after Israel launched large-scale strikes on Iran, stock futures fell, oil jumped, gold rallied, and the move was described as a flight-to-safety event.

[C] Reuters, Apr. 14, 2024: after Iran launched drones and missiles toward Israel, analysts highlighted higher oil, gold, and volatility as likely market reactions.

[D] Silver Institute, The Next Generation Metal, Dec. 2025 release: industrial demand for silver reached a record 680.5 million ounces in 2024, supported by electronics and electrical demand tied to AI-related applications.

[E] Semiconductor Industry Association, Feb. 6, 2026: global semiconductor sales increased 25.6 percent to \$791.7 billion in 2025 and were projected to reach roughly \$1 trillion in 2026.

[F] BlackRock IBIT product page, accessed Apr. 10, 2026: reported \$57.67 billion net assets, benchmark BRRNY, one holding, and 0.25 percent expense ratio.

[G] Strategy Inc. 8-K, filed Mar. 30, 2026: disclosed holdings of approximately 762,099 BTC as of Mar. 29, 2026.

[H] Coinbase shareholder letter / news release, Feb. 12, 2026: 2025 trading volume reached \$5.2 trillion, market share 6.4 percent, and more than 12 percent of all crypto in the world resided on Coinbase in 2025.

Final risk note

Cross-asset ratios are diagnostic tools, not certainty machines. They help you ask better questions about regime, leadership, and risk. They do not remove the need for position sizing, liquidity awareness, and scenario planning. Use them to improve judgment, not to eliminate uncertainty.